

ALP BARAN ERKUL

Boston, MA • (857) 335-7818 • abaran@bu.edu • <https://linkedin.com/in/alp-baran-erkul>

EDUCATION

Boston University, Boston, MA — B.S. in Mechanical Engineering

May 2027

2 Concentrations: Manufacturing Engineering, Technology Innovation | GPA: 3.75/4.00, Dean's List

Trustee Scholar Award: 1 of 15 students selected out of 80,495 applicants, and **the sole mechanical engineering recipient** for \$300,000 full merit-based award.

Selected member, Theta Tau Professional Co-Ed Engineering Fraternity. Participate in technical and professional development activities.

PROFESSIONAL EXPERIENCE

Elevaris Medical Devices, Wilmington, MA — Product Engineering Intern

May 2026 - Aug 2026

- Released **50+ SolidWorks CAD designs** and GD&T drawings for 9 OEM customers including Boston Scientific, Edwards, Cook Medical, Steris, and Avanos, applying tolerance stack-up analysis and DFM.
- Designed 8 different procedural needle geometries in CAD – lancet, quinke, tuohy, pencil-point - and various medical instruments from customer requirements through prototype and drawing release.
- Set nominal dimensions and tolerances under an active CAPA using 50+ measurements for each instrument and SPC/Cp/Cpk capability analysis, then released revised drawings through change control.
- Authored 3D scanning and inspection procedures adopted by Quality on Keyence VR6000 and VMS; executed ISO 13485 design V&V, IQ/OQ/PQ validation, and root cause analysis on nonconformances.

ASML, Hillsboro, OR — Engineering Project Coordinator Intern, Upgrades, Installations & Relocations

Jun 2025 - Sep 2025

- Coordinated 4 EUV lithography (EXE:5000/5200) installations **at Intel's largest U.S. fab**, aligning requirements, safety protocols, and milestones across cross-functional ASML (**Texas, Arizona, Netherlands**), **TRUMPF, Zeiss**, and customer teams.
- Exposed over 20% unused material across the majority of 25+ installation phases by running a Lean Six Sigma utilization study of 4 EUV installations, on parts costing \$1K to six figures each.
- Classified **500+ unused tools into 150+ usage benchmarks** from SAP ERP data, delivering a continuous improvement target list adopted by the next EUV installation program.
- Organized EUV installation plans and milestones at Intel, authored Best Known Methods, and presented weekly to program leads and managers across 3 U.S. sites which led my ideas to be utilized on ASML customer support sites like **Micron and TSMC**.

WeWalk Inc. — Mechanical Engineering Intern

Jun 2023 - Aug 2023

- Delivered CAD design for a lighter, thinner smart cane for visually impaired and low-vision users on a **two-time United Nations "Technology for a Better Life" honoree product line**.
- Raised predicted fatigue endurance 23% by redesigning the cane's curvature with FEA, then validated with the London team.

RESEARCH EXPERIENCE

Morphable Biorobotics Lab, Boston University (Prof. Ranzani) — Undergraduate Researcher

Dec 2024 - Present

- Advanced a **soft robotic platform for minimally invasive tricuspid valve replacement** in the beating heart by translating clinical and system-level constraints into mechanical design requirements and verification criteria.
- Designed and 3D-printed a motor driven syringe pump subsystem, integrating soft materials, sensors, and actuation into bench-tested prototypes.

ENGINEERING PROJECT EXPERIENCE & LEADERSHIP

- BAJA SAE, Manufacturing Team** Sep 2024 - Present
Manufactured off-road vehicle components by welding, machining, and soldering; designed for manufacturability in CAD.
- Mars Rover Club, Robotic Arm Hardware** Sep 2024 - May 2026
Designed, built, and tested a competition rover's robotic arm: SolidWorks CAD, FEA-driven iteration, 3D printing.
- Rocket Propulsion Group, Hybrid Engine Nozzle** Sep 2024 - Feb 2025
Sized throat and expansion geometry against target chamber pressure and thrust; built and tested the nozzle.
- Young Guru Academy Leadership School, Group Leader & Mentor** Apr 2020 - Present
Teach STEM to underprivileged children including refugees; selected leader of a mentor group, 6 years with the program.

PUBLICATIONS

- "The Role of Materials Science and Engineering in Medicine," ISSN 2822-6585
- "Computational Chemistry & Energy Structures," ISSN 2822-6593

TECHNICAL SKILLS

Mechanical Design & Analysis: SolidWorks, AutoCAD, ANSYS, Siemens NX, Onshape, Autodesk Fusion, COMSOL, 3D CAD modeling, 2D detail drawings, GD&T, tolerance stack-up analysis, FEA, DFM, mechanical design, materials selection, rapid prototyping.

Product & Systems Engineering: New product introduction (NPI), design requirements, concept selection, design reviews, ISO 13485 design controls, verification and validation (V&V), FMEA, risk management, IQ/OQ/PQ process validation, test protocols and reports.

Quality Engineering & Process Improvement: SPC, process capability (Cp/Cpk), Lean Six Sigma, continuous improvement, CAPA, root cause analysis, nonconformance investigation, drawing release, inspection procedure and revision control.

Manufacturing Engineering & Metrology: Keyence VR6000 3D scanning and VMS, dimensional inspection, CNC milling and turning, injection molding, welding, casting, laser cutting, waterjet, additive manufacturing (FDM), 3D printing.

Programming & Data: Python, MATLAB, Arduino, SAP ERP, Excel, Power BI, Tableau.